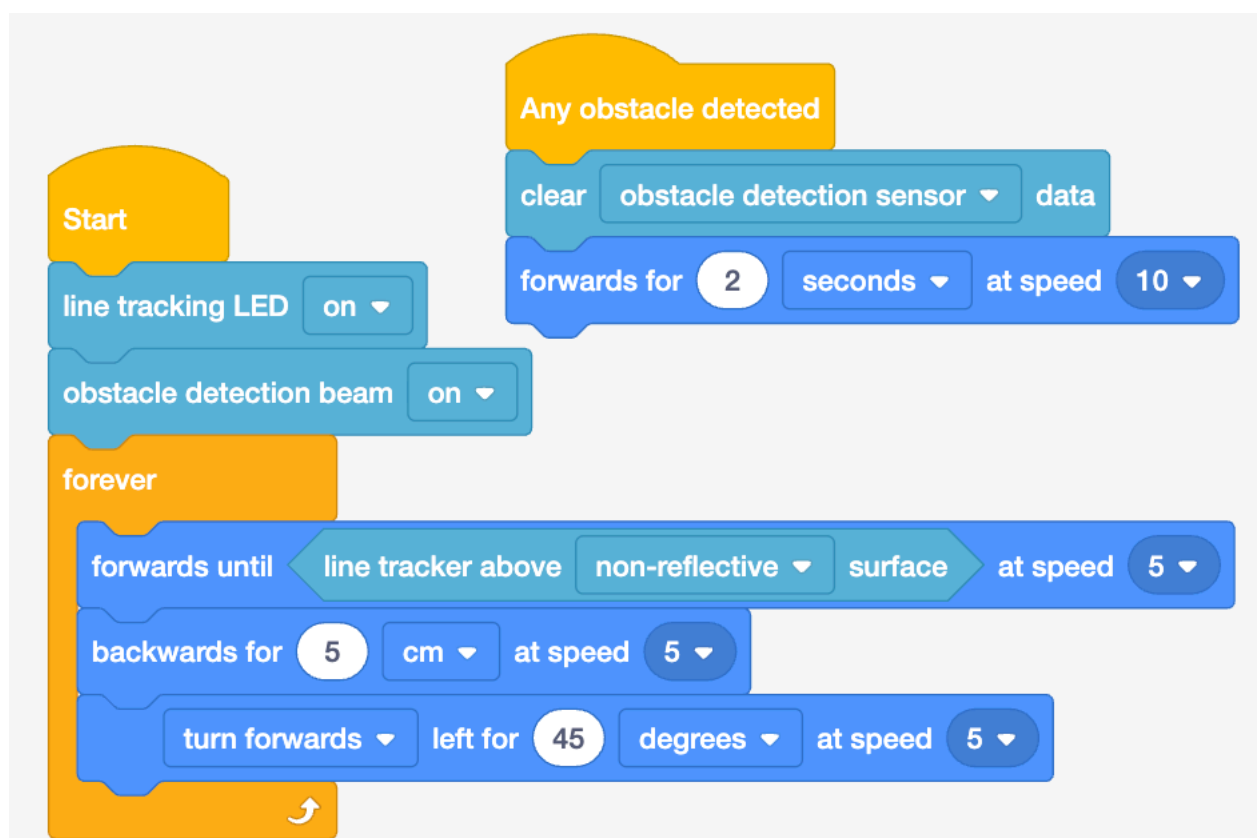


This code programs your robot to act as an automated sumo wrestler with a "search, evade, and attack" routine:

- **Initialization (Start):** Powers on the line-tracking LEDs and the obstacle-detection infrared beam so the robot's sensors are active the moment you hit run.
- **The Roam & Boundary Check (forever):**
 1. Drives forward slowly (speed 5) across the ring.
 2. Stops the moment the line tracker detects a **reflective** surface (the ring's boundary).
 3. Reverses 5 cm away from the edge.
 4. Turns left by 45 degrees to aim itself back toward the center of the ring, then repeats.
- **The Attack Routine (Any obstacle detected):** Runs independently as an interrupt. The second the obstacle sensor spots another robot or object, it clears the sensor data and immediately charges forward at top speed (speed 10) for 2 seconds to try to push the opponent out of the ring.



The "Robot Rumble" program takes last week's "Bounce In Borders" code and upgrades it with a **combat behavior layer**.

Here is how they compare:

- **The Boundary Defense is Identical:** Both programs use the same core loop to roam the ring, detect the border line, back up 5 cm, and turn 45 degrees to stay inside.
- **Added Sensor Setup:** The Robot Rumble **Start** block turns on both the *line-tracking LEDs* and the *obstacle-detection beam*, whereas last week's program only needed the line tracker.
- **The Attack Routine (The Big Difference):** Robot Rumble introduces a separate event block (**Any obstacle detected**) that runs simultaneously. The moment an opponent or object enters its sensor view, it ignores its normal wandering loop and instantly charges forward at top speed for 2 seconds to push them out.

Bounce In Borders

Solution

